GOOD MAPS WITH SURJECTION AND VANISHING HOM

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Allowed standard prerequisites (editorial).

- Enough negative objects. definition-level assumption of the statement. Provides N with the surjection and vanishing Hom.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough negative objects. Let $X \in D^b(\mathcal{A})$. Let $b \in \mathbf{Z}$ with $H^i(X) = 0$ for $i > b$. Then there exists a map $N[-b] \to X$ such that the induced map $N \to H^b(X)$ is surjective and* $\mathrm{Hom}(H^b(X), N) = 0$.

Proof. Using the truncation functors we can represent X by a complex $A^a \to A^{a+1} \to \ldots \to A^b$ of objects of $\mathcal{A}$. Choose N in $\mathcal{A}$ such that there exists a surjection $t : N \to A^b$ and such that $\mathrm{Hom}(A^b, N) = 0$. Then the surjection t defines a map $N[-b] \to X$ as desired. □

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